



JAIN
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SCHOOL OF
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DEPARTMENT OF CS & IT

Programme: MCA

Activity - 1
Cloud Analytics
25MCASC302

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1. Introduction

This document accompanies Activity 1 (Portfolio Creation) and describes how my personal/professional cloud engineering portfolio was designed, developed, and deployed. The portfolio is a fully static, responsive single-page site built to showcase my cloud engineering background — AWS, Azure, DevOps, and Kubernetes projects — completed during my MCA studies and my Cloud Computing internship at Codetech.

The site is hosted on Microsoft Azure Static Web Apps and the source code is version-controlled and maintained in a public GitHub repository. The sections below cover the design approach, the technology stack, key features, the deployment process, and screenshots of the live site.

GitHub Repository	github.com/prem25005322-coder/prem-portfolio
Live Portfolio Link	calm-river-0a673ff00.7.azurestaticapps.net

2. Design and Development Approach

2.1 Concept

- Designed the portfolio around a “cloud engineer” theme rather than a generic resume layout — using a dark, terminal/CLI-inspired visual style to reflect an infrastructure and DevOps identity.
- Structured the page as a single-page application with anchor-linked navigation: Home, Engineering, Architecture & Projects, Ecosystem (Skills), Trajectory (Experience), Certifications, and Contact.
- Prioritized a content hierarchy that leads with cloud architecture projects first, since that is the strongest signal of hands-on capability.

2.2 Development Process

- Hand-coded the site using plain HTML5, CSS3, and vanilla JavaScript — no frontend framework — to keep the deployment fully static and framework-free.
- Built custom CSS components for the animated “live cloud stack” pipeline diagram, project filter tabs (All Systems / AWS & Serverless / DevOps & IaC / Azure & Data / Mobile & Full Stack), and the skills grid.
- Implemented an interactive CLI/terminal widget on the page (commands such as whoami, skills, projects, aws status, azure lake, certs, cat resume) using JavaScript to simulate shell output — reinforcing the cloud/DevOps theme.
- Used semantic HTML and metadata (title, description, keywords, Open Graph tags) for SEO and link-preview quality.
- Tested responsiveness across breakpoints so the layout, navigation, and terminal widget remain usable on mobile and desktop.

2.3 Content Strategy

- Organized experience into four categories on the Projects section: AWS & Serverless, DevOps & IaC, Azure & Data, and Mobile & Full Stack, so a recruiter can filter by area of interest.

- Each project card follows a consistent format: architecture flow (e.g. S3 → Lambda → DynamoDB), a short description, 1–2 concrete implementation bullet points, and technology tags.
- Included a dedicated Certifications section and an Experience & Trajectory timeline covering the Codetech internship, prior internships, and academic milestones (MCA and BCA).

3. Technologies, Tools, Frameworks and Platforms Used

3.1 Frontend

- HTML5 – semantic page structure and content markup
- CSS3 (vanilla, custom stylesheet) – layout, theming, responsive design, animations
- Vanilla JavaScript (ES6+) – navigation behavior, project filtering, and the interactive CLI/terminal widget

3.2 Hosting & Deployment

- Microsoft Azure Static Web Apps – hosting platform for the deployed site
- Deployment carried out directly through the Azure Portal, linking the Static Web App resource to the GitHub repository as the content source
- Azure-managed SSL/HTTPS and a default *.azurestaticapps.net domain for the live link

3.3 Version Control

- Git – source control for the project
- GitHub – remote repository hosting, used as the content source for the Azure Static Web App

3.4 Design/Productivity Aids

- Generative AI tools – used to accelerate drafting of content copy, validate CSS/JS snippets, and review the documentation structure

4. Key Features and Functionalities

- Responsive single-page layout with sticky anchor navigation (Home, Engineering, Projects, Skills, Experience, Certs, Contact).
- Animated “live cloud stack” visual on the hero section illustrating a serverless pipeline (S3 → Lambda → DynamoDB, with status indicators such as “200 OK” and “<18ms”).
- Interactive CLI/terminal widget that responds to typed or clicked commands (whoami, skills, projects, aws status, azure lake, certs, cat resume, sudo hire, clear).
- Filterable project showcase across 8 projects and 4 categories (All Systems, AWS & Serverless, DevOps & IaC, Azure & Data, Mobile & Full Stack), each with an architecture-flow summary and GitHub source link.
- Technical ecosystem grid grouping skills into six categories: AWS Cloud Platforms, Microsoft Azure, IaC & DevOps, Programming Languages, Observability & QA, and AI & Acceleration.
- Experience & Trajectory timeline covering the Codetech internship, two earlier internships, an academic research milestone (Azure Data Lake), and MCA/BCA education.
- Verified Certifications section (AWS Cloud Architecting, AWS Networking, Generative AI, Cybersecurity Fundamentals).

- Contact section with direct email, phone/WhatsApp, location, LinkedIn and GitHub links, plus a message/contact form.

5. Screenshots

The screenshots below correspond to the sections of the deployed portfolio.

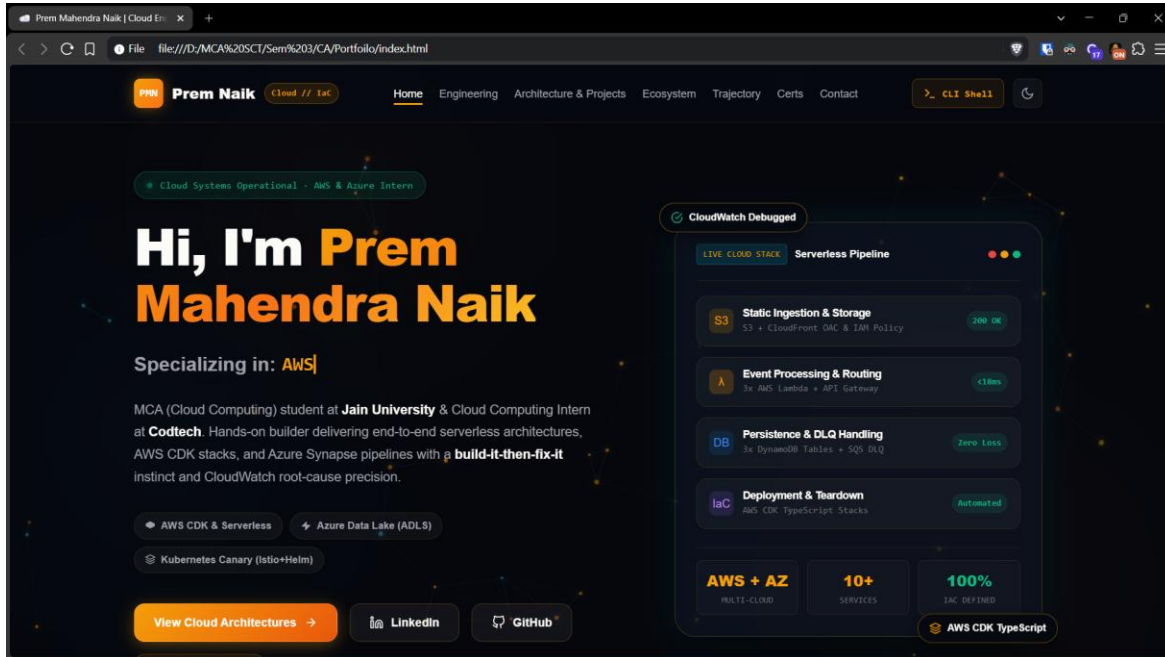


Figure 1: Home / hero section with the animated live cloud stack pipeline

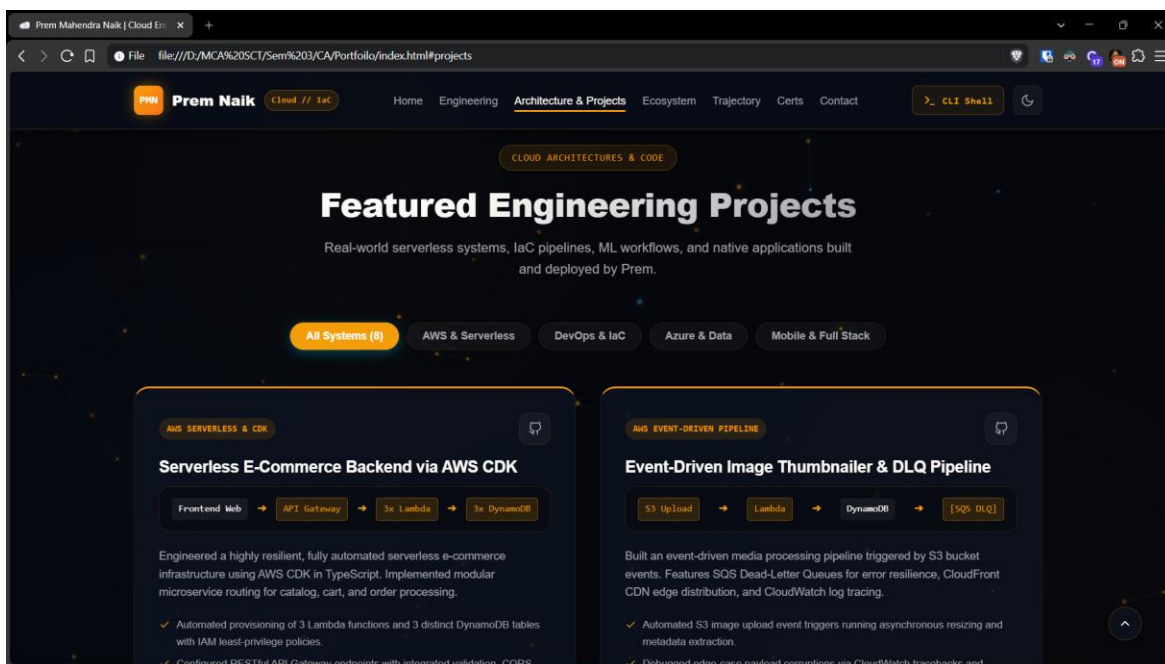


Figure 2: Architecture & Projects section with category filter tabs

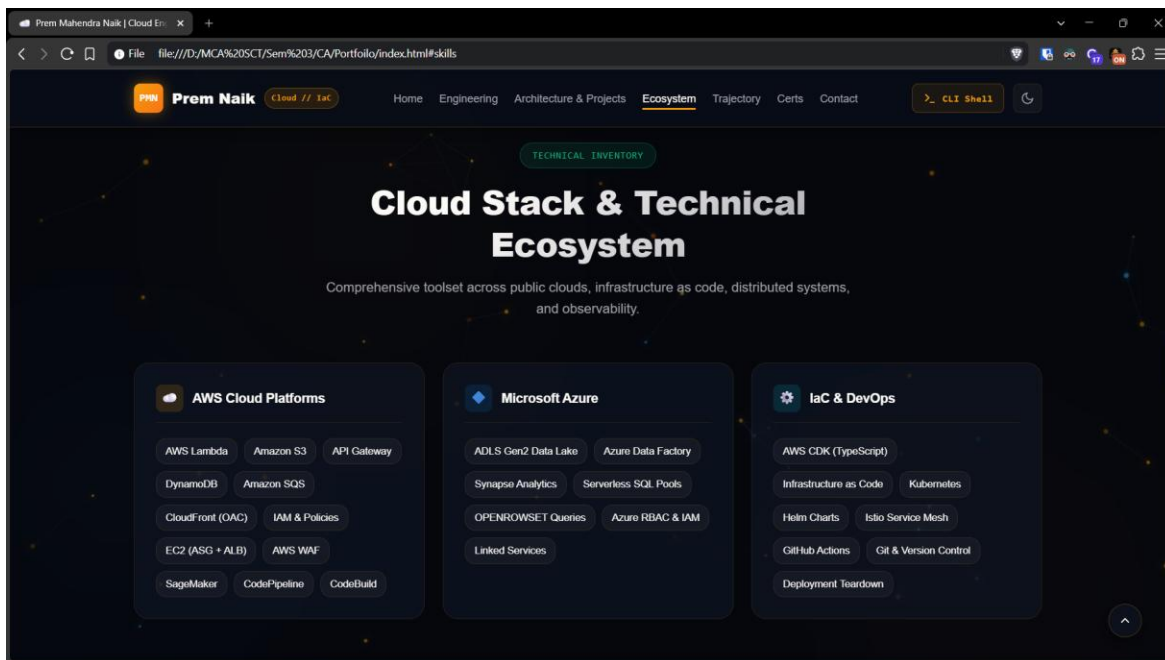


Figure 3: Technical Ecosystem / skills grid

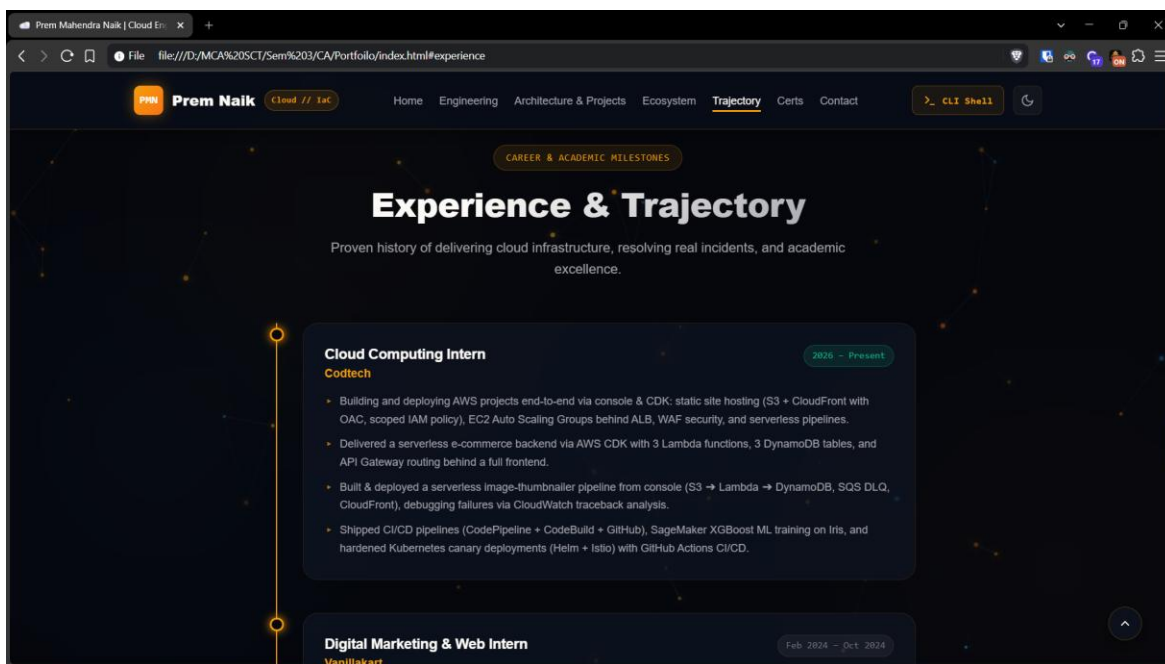


Figure 4: Experience & Trajectory timeline

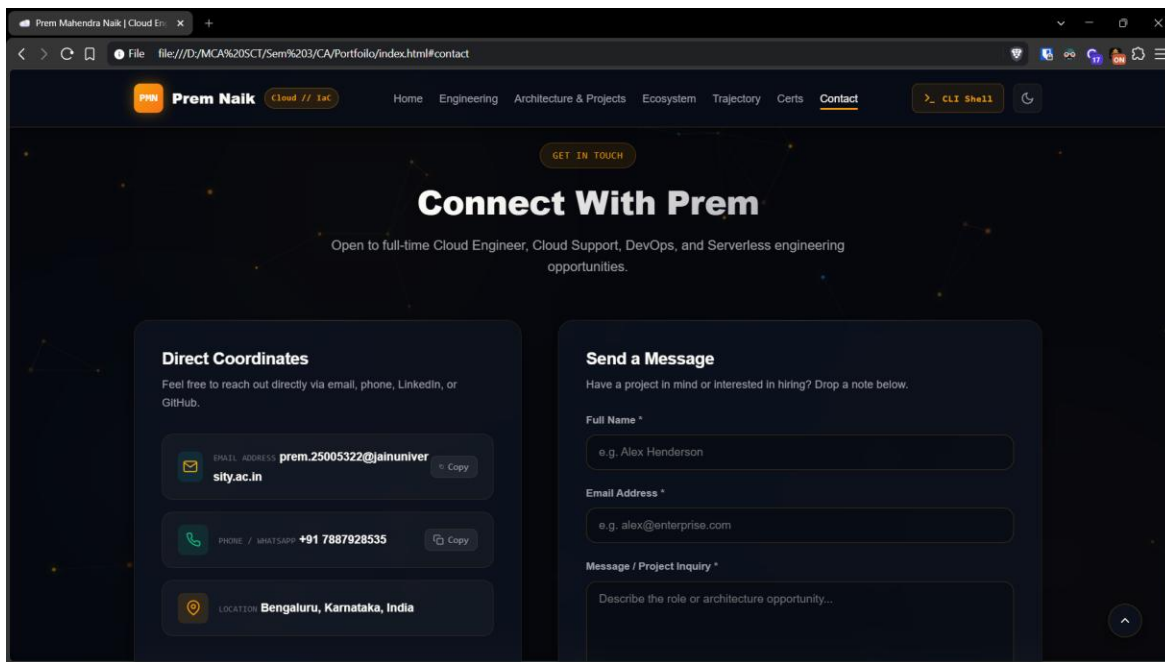


Figure 5: Contact section

6. Design and Implementation Approach — Summary

- Design goal: make the portfolio itself read as a cloud-engineering artifact — an animated pipeline diagram and a working terminal widget double as informal proof of frontend/JS capability, alongside the listed backend/cloud projects.
- Implementation choice: kept the stack framework-free (plain HTML/CSS/JS) so the site deploys as pure static content with zero build step, which pairs naturally with Azure Static Web Apps.
- Content over decoration: each project card intentionally leads with the architecture flow and one or two concrete technical bullet points rather than long prose, to keep the page scannable.
- Accessibility/perf: static assets only (no client-side framework runtime to download), semantic HTML, and a metadata block for SEO and social-link previews.

7. Deployment Process

- Pushed the finished HTML/CSS/JS project to the GitHub repository: prem25005322-coder/prem-portfolio.
- Created an Azure Static Web App resource directly through the Azure Portal.
- Linked the Static Web App to the GitHub repository and configured the app/build output paths for the static site.
- Azure provisioned the deployment and issued the live default hostname: calm-river-0a673ff00.7.azurestaticapps.net.
- Verified the deployed site loads correctly, all anchor navigation works, and the project/skills/contact sections render as expected.

8. Conclusion

The portfolio successfully consolidates my AWS, Azure, DevOps, and mobile/full-stack project work into a single deployed, publicly accessible site, backed by a version-controlled GitHub repository. It satisfies the Activity 1 requirements: the project and documentation are maintained in GitHub, the site is deployed with a working live link on Azure Static Web Apps, and both links are recorded below for submission.

GitHub Repository: <https://github.com/prem25005322-coder/prem-portfolio>

Live Portfolio Link: <https://calm-river-0a673ff00.7.azurestaticapps.net/>